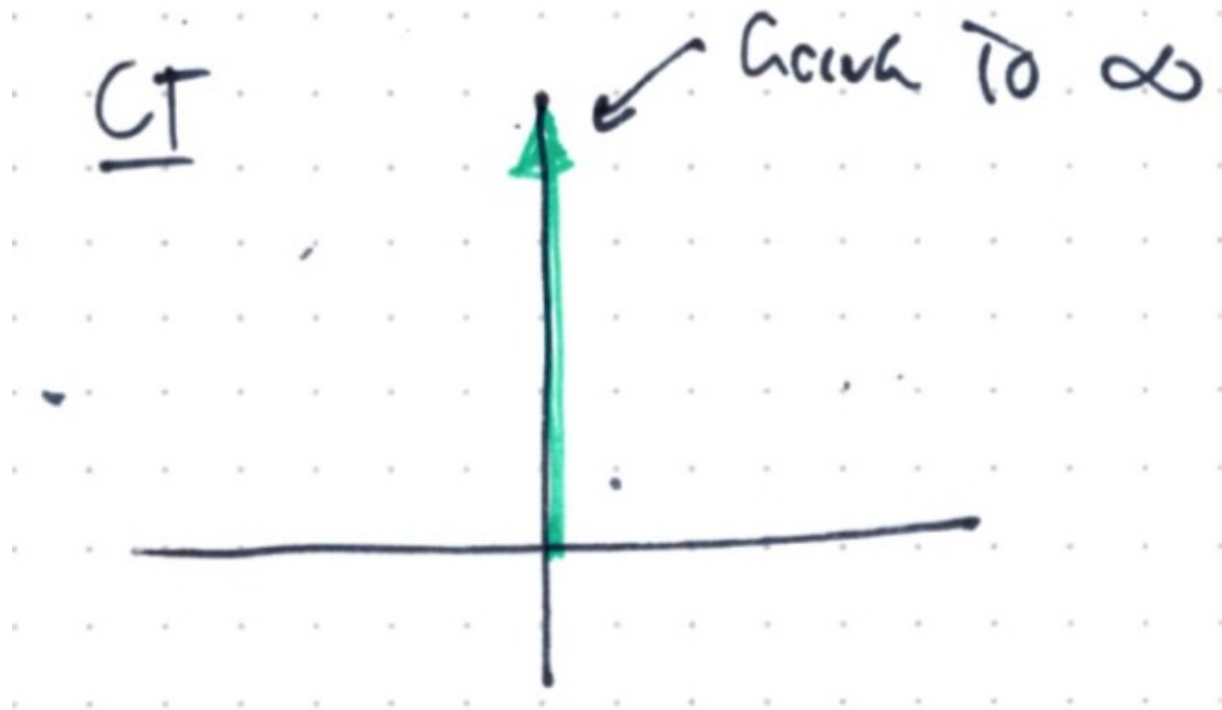
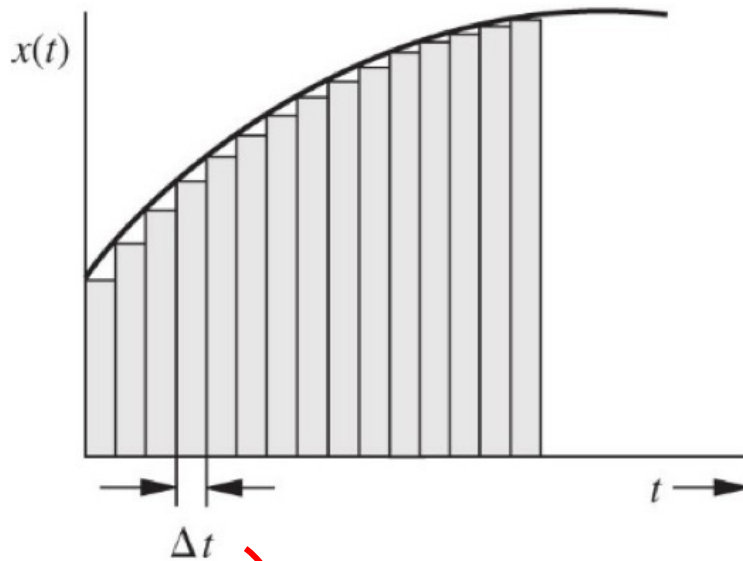




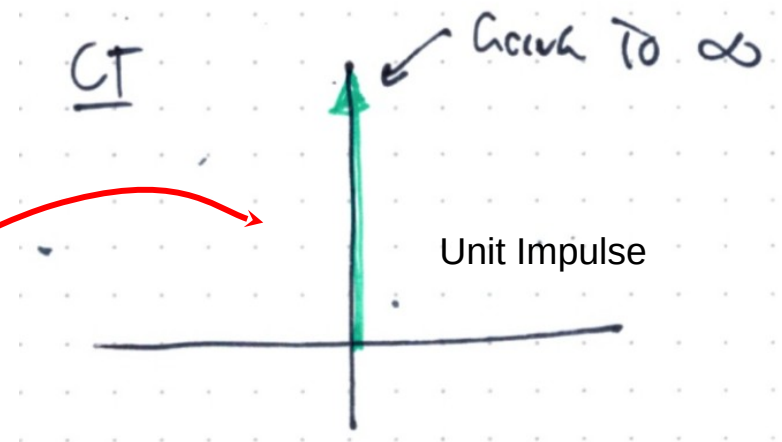
SYDE252 - W05A – Impulse Response
Dr. Matt Borland

Additivity and Scaling Properties of Linear Systems

For linear systems we know we can break up the input signal into a bunch of parts and send them through the system, then add them up on the other side...



As Δt goes to 0, this becomes similar to our unit impulse we defined before.

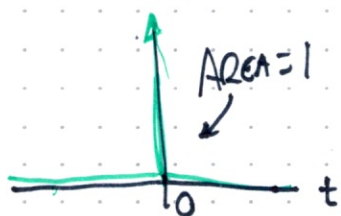


THE UNIT IMPULSE RESPONSE: $h(t)$

①

AS ALWAYS $x(t) \rightarrow [A] \rightarrow y(t)$

FOR A SPECIAL INPUT $\delta(t)$ WE HAVE A SPECIAL OUTPUT. FOR LTI SYSTEMS



$$\delta(t) \rightarrow [A] \rightarrow h(t)$$

UNIT IMPULSE



UNIT IMPULSE RESPONSE

WITH IC's = 0

REMEMBER FOR LTIC SYSTEMS

$$a_1 x_1(t) \rightarrow a_1 y_1(t)$$

$$a_2 x_2(t) \rightarrow a_2 y_2(t)$$

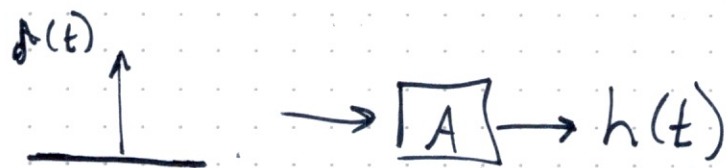
$$+ \vdots$$
$$\frac{x(t)}{\quad} \rightarrow \frac{y(t)}{\quad}$$

SAMPLING PROPERTY

$$\int_{-\infty}^{\infty} \delta(t) \cdot f(t) dt = f(0)$$

IF WE CAN BREAK $x(t)$ INTO A BUNCH OF $\delta(t)$ 'S
THINGS $\rightarrow$ THEN WE CAN JUST ADD UP A BUNCH
OF $h(t)$ 'S TO GET THE OUTPUT

IF WE KNOW $h(t)$ WE CAN SOLVE FOR
ANY $y(t)$ FOR ANY $x(t)$ WE WANT TO PUT IN!



$\delta(t)$ APPLIED AT $t=0$ WITH ALL INITIAL CONDITIONS = ZERO AT $t=0^-$ ← ZERO FROM THE LEFT SIDE

AT $t=0$ THE $\delta(t)$ STORES ENERGY IN THE SYSTEM INSTANTANEOUSLY. AT $t=0^+$

AT $t>0$ $\delta(t)=0$ SO THERE IS NO MORE INPUT AFTER WE PUT IN THE IMPULSE

WE SAW THAT FOR ZERO INPUT $x(t)=0$ WE HAVE A RESPONSE DEFINED BY THE MODES

$$h(t) = \text{MODE TERMS} \quad t > 0^+$$

WHAT HAPPENS AT $t=0$?

SINCE $t=0$ IS ONLY A SINGLE SLICE OF TIME

AT MOST
$$h(t) = A_0 \delta(t) + \text{MODE TERMS}$$

LTIC SYSTEM DESCRIBED

3

$$(D^N + a_1 D^{N-1} + \dots + a_{N-1} D + a_N) y(t) = (b_{N-M} D^M + b_{N-M+1} D^{M-1} + \dots + b_{N-1} D + b_N) x(t)$$

WE SAID WE ARE ONLY CONCERNED WITH SYSTEMS WHERE $M \leq N$... SO LET $M=N$ IN THE MOST GENERAL CASE

$$(D^N + a_1 D^{N-1} + \dots + a_{N-1} D + a_N) y(t) = (b_0 D^N + b_1 D^{N-1} + \dots + b_{N-1} D + b_N) x(t)$$

IF SET $x(t) = \delta(t)$ AND $y(t) = h(t) = A_0 \delta(t) + \text{modes}$

$$(D^N + a_1 D^{N-1} + \dots + a_{N-1} D + a_N) (A_0 \delta(t) + \text{modes}) = (b_0 D^N + b_1 D^{N-1} + \dots + b_{N-1} D + b_N) \delta(t)$$

THESE HAVE TO MATCH BECAUSE OF THE ORDER OF THEIR DERIVATIVE

$$\therefore A_0 = b_0 \Rightarrow h(t) = b_0 \delta(t) + \text{modes} \} \text{ IF } M=N$$

$$b_0 = 0 \text{ IF } M < N \quad h(t) = \text{modes} \} \text{ IF } M < N$$

TO DETERMINE THE COEFFICIENTS OF THE MODES WE USE "IMPULSE MATCHING"

(4)

For A LTIC SYSTEM:

$$\begin{aligned} & (D^N + a_1 D^{N-1} + \dots + a_{N-1} D + a_N) y(t) \\ &= (b_0 D^N + b_1 D^{N-1} + \dots + b_{N-1} D + b_N) x(t) \end{aligned}$$

From our proof (see 2.8):

$$h(t) = b_0 \delta(t) + [P(D) y_n(t)] u(t)$$

↑
IS A LINEAR COMBINATION
OF THE MODES

With the I.C.'s

$$y_n(0) = \dot{y}_n(0) = \dots = y_n^{(N-2)}(0) = 0$$

$$\text{AND } y_n^{(N-1)}(0) = 1$$

$$N=1: \quad y_n(0) = 1$$

$$N=2: \quad y_n(0) = 0, \quad \dot{y}_n(0) = 1$$

$$N=3: \quad y_n(0) = \dot{y}_n(0) = 0, \quad \ddot{y}_n(0) = 1$$

DETERMINE $x(t)$ FOR A SYSTEM SPECIFIED BY ⑤

$$(D^2 + 3D + 2)y(t) = Dx(t)$$

$$\frac{d^2 y(t)}{dt^2} + 3 \frac{dy(t)}{dt} + 2y(t) = \frac{dx(t)}{dt} \quad M < N \text{ NICE!}$$

WHAT IS OUR N ? $N=2$

CHARACTERISTIC POLYNOMIAL

$$(\lambda^2 + 3\lambda + 2) = (\lambda + 1)(\lambda + 2)$$

CHAR. ROOTS ARE $\lambda = -1$ $\lambda = -2$

$$y_h(t) = C_1 e^{-t} + C_2 e^{-2t}$$

WE STILL NEED TO SOLVE FOR C_1 & C_2

FROM IMPULSE FOR SYSTEM $N=2$

$$\dot{y}_h(0) = 1 \quad y_h(0) = 0$$

$$\text{FIND } y_h(t) \Rightarrow \dot{y}_h(t) = -C_1 e^{-t} - 2C_2 e^{-2t}$$

SO SET $t=0$ AND SOLVE FOR C_1 & C_2

$$\begin{cases} 0 = C_1 + C_2 \\ 1 = -C_1 - 2C_2 \end{cases} \text{ 2 EQNS 2 UNKNOWNNS}$$

$$C_2 = -1 \quad C_1 = 1$$

$$\Rightarrow y_h(t) = e^{-t} - e^{-2t}$$

(6)

SO WE HAVE A SOLUTION FOR $y_n(t)$, THE MODES,

BUT WE WANT

$$h(t) = (P(D) y_n(t)) u(t)$$

↑
DERIVATIVE
OPERATOR

↑
MODES

↑
UNIT
STEP

$$y_n(t) = e^{-t} - e^{-2t}$$

AND REMEMBER:

$$Q(D) y(t) = P(D) x(t)$$

FOR OUR SYSTEM:

$$\underbrace{(D^2 + 3D + 2)}_{Q(D)} y(t) = \underbrace{(D)}_{P(D)} x(t)$$

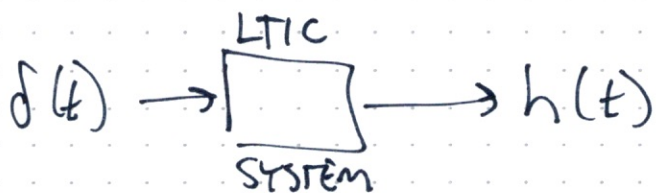
SO $P(D) = D$ ← JUST MEANS TAKE $\frac{d}{dt}$ OF WHATEVER IT IS OPERATING ON

$$P(D) y_n(t) = D y_n(t) = \frac{d y_n(t)}{dt} = -e^{-t} + 2e^{-2t}$$

SO PUTTING IT ALL TOGETHER:

$$h(t) = (-e^{-t} + 2e^{-2t}) u(t)$$

WHERE



SO WHY DO ALL OF THIS? BECAUSE:

$$y(t) = x(t) * h(t)$$

FOR LTI SYSTEMS

IF WE KNOW $h(t)$, WE CAN INPUT ANY $x(t)$ WE WANT AND CAN CALCULATE THE OUTPUT $y(t)$!

EXAMPLE Find $h(t)$ for (2.3-5)

$$\underbrace{(D^2 + 6D + 9)}_{Q(D)} y(t) = \underbrace{(2D + 9)}_{P(D)} x(t)$$

$$y(t) = P(D) x(t)$$

CHARACTERISTIC
POLYNOMIAL:

$$(\lambda^2 + 6\lambda + 9)$$

$$= (\lambda + 3)(\lambda + 3)$$

ORDER OF THE DE: $N=2$

REPEATED ROOTS, so

$$y_h(t) = C_1 e^{-3t} + C_2 t e^{-3t}$$

NEXT STEP: SOLVE FOR THE DERIVATIVES $\rightarrow N=2$ so we
NEED $y_h(t), \dot{y}_h(t)$

From the proof SEE 2.8

$$\begin{cases} y_h(0) = 0 \leftarrow \text{THE OTHER DERIVATIVES} = 0 \\ \dot{y}_h(0) = 1 \leftarrow \text{THE HIGHEST ORDER DERIVATIVE} = 1 \end{cases}$$

$$\dot{y}_h(t) = -3C_1 e^{-3t} + C_2 e^{-3t} - 3C_2 t e^{-3t}$$

$$y_h(t) = C_1 e^{-3t} + C_2 t e^{-3t}$$

$\rightarrow$ SUB-IN AND SOLVE FOR $t=0$

$$1 = -3C_1 + C_2 \Rightarrow C_1 = 0, C_2 = 1$$

$$0 = C_1$$

WE'VE SOLVED FOR OUR CONSTANTS... SO:

$$y_h(t) = t e^{-3t}$$

$$P(D) = 2D + 9 \Rightarrow P(D) y_h(t) = 2D(t e^{-3t}) + 9 t e^{-3t}$$

$$= 2(e^{-3t} - 3t e^{-3t}) + 9 t e^{-3t}$$

$$= 2e^{-3t} + 3t e^{-3t}$$

$$h(t) = [P(D) y_h(t)] u(t)$$

$$\therefore h(t) = (2e^{-3t} + 3t e^{-3t}) u(t)$$

WE KNOW:

$$h(t) = [P(D) y_h(t)] u(t)$$

$$\delta(t) \rightarrow \boxed{A} \rightarrow h(t)$$

(7)